

Project 4: Preference Elicitation

Human-Centric Artificial Intelligence

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1. Introduction

This report describes the design and implementation of a movie preference elicitation system and a user study comparing two elicitation interfaces. The goal is to develop a movie recommender that quickly adapts to a new user's preferences through a limited number of interactions. We use the IMDB 5000 Movie Dataset (4,998 movies, 160 features after preprocessing) and model user preferences using the Bradley–Terry framework extended to rankings.

2. Task 1: Feature Representation

2.1 Feature choice

A good feature representation for a movie recommender must capture the attributes users actually care about when deciding what to watch. We chose five categories of features:

- Genre (one-hot encoded, 26 genres): the most decisive factor in movie preference — a user who enjoys action films will consistently prefer action over romance regardless of other attributes.
- Country of production (one-hot encoded): captures regional filmmaking styles and cultural sensibilities that influence viewer preference.
- Language (one-hot encoded): directly affects accessibility and preference, particularly for users who prefer or avoid subtitled content.
- Content rating (one-hot encoded): reflects maturity and tone — a user preferring family-friendly films is systematically different from one who prefers R-rated content.
- Duration (normalized): captures preference for short vs. long films — some users consistently prefer concise films, others prefer epic runtimes.
- Release year (normalized): captures era preference — some users prefer classic films, others strongly prefer contemporary releases.

Features were extracted using MultiLabelBinarizer for genres and countries (since movies can belong to multiple), OneHotEncoder for language and content rating, and StandardScaler for the two continuous features (duration and year). Missing numerical values were imputed with the column median; missing categorical values were filled with 'Unknown'.

2.2 Feature matrix

Property	Value
Number of movies	4,998
Total features	160
Genre features	26 (one-hot)
Country features	66 (one-hot)
Language features	49 (one-hot)
Content rating features	17 (one-hot)
Numerical features	2 (duration, year — normalized)

The utility of movie x for user with preference vector w is defined as $U(x) = w^T x$, where $x \in \mathbb{R}^{160}$ is the feature vector and $w \in \mathbb{R}^{160}$ is the latent user preference vector estimated from interactions.

3. Task 2: Extended Bradley–Terry Model

3.1 Standard Bradley–Terry model

The standard Bradley–Terry model defines the probability that item i is preferred to item j in a pairwise comparison as:

$$P(i \succ j) = \exp(U(i)) / (\exp(U(i)) + \exp(U(j)))$$

where $U(i) = w^T x_i$ is the utility of item i . This models pairwise preferences but does not directly handle rankings of more than two items.

3.2 Extension to rankings: the Plackett–Luce model

The standard extension of Bradley–Terry to full rankings of n items is the Plackett–Luce model. The probability of a full ranking $i_1 \succ i_2 \succ \dots \succ i_n$ is:

$$P(i_1 \succ i_2 \succ \dots \succ i_n) = \prod_k \exp(U(i_k)) / \sum_{j \geq k} \exp(U(i_j))$$

Intuitively, at each position k , the probability of selecting item i_k from the remaining items is proportional to its exponentiated utility relative to all items not yet placed. This is a natural generalization: at each step, the most preferred remaining item is chosen according to the same logistic rule as in standard Bradley–Terry.

3.3 Justification

The Plackett–Luce model is the unique extension of Bradley–Terry to rankings satisfying Luce's choice axiom (independence of irrelevant alternatives). It reduces exactly to Bradley–Terry when $n = 2$, ensuring consistency between the two elicitation designs. The log-likelihood for a ranking is:

$$\ell(\theta) = \sum_k [U(i_k) - \log(\sum_{j \geq k} \exp(U(i_j)))]$$

Preference vectors w are estimated by maximizing this log-likelihood (for rankings) or the standard pairwise log-likelihood (for comparisons) using L-BFGS-B optimization, implemented in SciPy.

4. Task 3: User Study Design

4.1 Research hypothesis

We test the following directional hypothesis:

H_1 : Ranking-based elicitation (Design 2) produces higher-quality personalized recommendations than pairwise comparison (Design 1) when the number of interactions is matched.

The null hypothesis H_0 is that there is no significant difference in recommendation quality between the two designs.

4.2 Study design

We use a between-subjects design: each participant is randomly assigned to either Design 1 or Design 2 and completes only that condition. This avoids order and learning effects that would arise from a within-subjects design.

Property	Design 1 (Pairwise)	Design 2 (Ranking)
Task	Choose between 2 movies	Rank 10 movies
Rounds	10 pairwise comparisons	2 ranking tasks
Movies seen	20 unique movies	20 unique movies

Movie selection	Uniform random	Uniform random
Est. time	5–8 minutes	5–8 minutes

Both designs expose participants to the same number of unique movies (20), ensuring the comparison is fair in terms of information available to the preference estimator.

4.3 Metrics

Three primary metrics are collected:

- Recommendation quality: after elicitation, 5 movies are recommended and the participant rates each on a 1–5 star scale. The mean rating across 5 recommendations is the primary outcome measure.
- User experience: a post-task questionnaire measures perceived ease of the task (1–5), satisfaction with recommendations (1–5), and confidence that recommendations reflect actual preferences (1–5).
- Efficiency: the number of interactions required for the preference estimate to stabilize, measured by the cosine similarity between w at round k and w at round $k+1$ (computed offline from logged interaction data).

4.4 Participants and recruitment

Target sample: 30 participants per condition (60 total), sufficient for 80% statistical power at $\alpha = 0.05$ for a medium effect size (Cohen's $d = 0.5$) using a two-sample Mann–Whitney U test.

Inclusion criteria: 18 years of age or older; watches at least one movie per month. Exclusion criteria: prior knowledge of the study design or the recommender system.

Recruitment: university participant pool or online crowdsourcing platforms (e.g. Prolific). Participants are compensated according to platform standards. They are not informed of which design they are assigned to.

4.5 Procedure

Each participant follows this sequence:

- Read the participant information sheet and provide informed consent via the consent form.
- Read the task instructions (tailored to their assigned design).
- Complete the elicitation phase (10 pairwise comparisons or 2 ranking tasks of 10 movies).
- View 5 personalized movie recommendations and rate each on a 1–5 star scale.
- Complete the post-task questionnaire (ease, satisfaction, confidence).
- Read the debrief explaining the study purpose.

Estimated total time: 5–10 minutes. The interface requires no researcher presence — participants complete the study fully independently via the web interface.

4.6 Statistical analysis

The primary analysis compares mean recommendation ratings between Design 1 and Design 2 using a two-sample Mann–Whitney U test (non-parametric, since Likert-scale ratings are ordinal). Effect size is reported as rank-biserial correlation. Secondary analyses compare user experience scores (ease, satisfaction, confidence) using the same test. A Bonferroni correction is applied for the three secondary comparisons ($\alpha' = 0.05/3 \approx 0.017$).

We also plot recommendation quality as a function of the number of interactions for both designs (using cosine similarity of successive w estimates) to characterize efficiency.

4.7 Ethical considerations

The study involves no deception, collects no personally identifiable information, and carries minimal risk. Participation is voluntary and participants may withdraw at any time without consequence. The consent form clearly states these rights. Data is stored anonymously and used solely for academic research purposes.

5. Conclusion

This report described a complete movie preference elicitation system built on the IMDB 5000 dataset. A 160-dimensional feature representation captures genre, country, language, content rating, duration, and release year. The Bradley–Terry model was extended to rankings using the Plackett–Luce formulation, which is the principled and unique generalization satisfying Luce's choice axiom. A rigorous between-subjects user study was designed to test whether ranking-based elicitation produces better recommendations than pairwise comparison when the number of exposed movies is matched, with clear metrics, a defined recruitment process, a complete participant procedure, and a pre-specified statistical analysis plan.